

Module 4: ELB Assignment

Problem Statement:

“You work for XYZ Corporation that uses on premise solutions and some limited number of systems. With the increase in requests in their application, the load also increases. So, to handle the load the corporation has to buy more systems almost on a regular basis. Realizing the need to cut down the expenses on systems, they decided to move their infrastructure to AWS”.

Tasks To Be Performed:

1. To create a Classic Load Balancer and register 3 EC2 instances with different web pages running in them.
2. To migrate the Classic Load Balancer into an application load balancer.

Structure Flow:

3 EC2 instances → Apache → 3 different webpages
↓
Classic Load Balancer → test
↓
migrate to Application Load Balancer → test again.

1. Created 3 EC2 instances

EC2 → Instances → Launch instance

Selected Configuration and launched 3 EC2 servers.

ELB-Server-1, ELB-Server-2, ELB-Server-3

EC2 > Instances > Launch an instance

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

ELB-Server-1

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

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Ubuntu

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Red Hat

SUSE Linux

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debian

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Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI
ami-0fef201115eefe936 (64-bit (x86), uefi-preferred) / ami-0eb45f74aa8a20238 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Summary

Number of instances

3

When launching more than 1 instance, consider [EC2 Auto Scaling](#)

Software Image (AMI)

Amazon Linux 2023 AMI 2023.12.....[read more](#)
ami-0fef201115eefe936

Virtual server type (instance type)

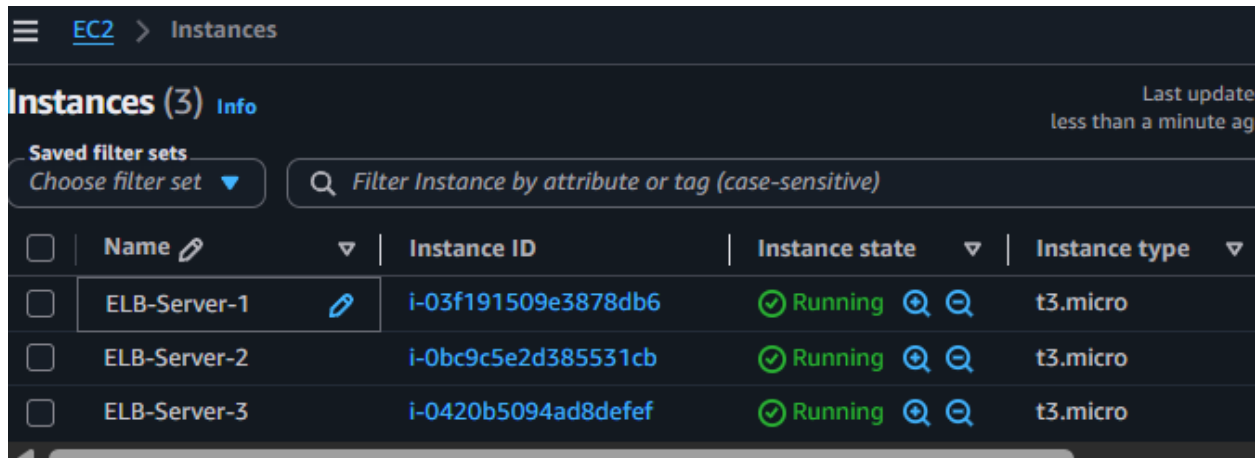
t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB



```
#_
~\##### Amazon Linux 2023
~\#####\
~~\###|
~~\#/ https://aws.amazon.com/linux/amazon-linux-2023
~~v-' '->
~~~~
~~~~.-.
~/m/' -/
```

```
[ec2-user@ip-172-31-11-238 ~]$ sudo dnf update -y
sudo dnf install httpd -y
sudo systemctl enable --now httpd
```

```
Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64      apr-util-1.6.5-1.amzn2023.0.1.x86_64      apr-util-ldap-1.6.5-1.amzn2023.0.1.x86_64
apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64      generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      httpd-2.4.68-1.amzn2023.0.1.x86_64
httpd-core-2.4.68-1.amzn2023.0.1.x86_64      httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch      httpd-tools-2.4.68-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64      mailcap-2.1.49-3.amzn2023.0.3.noarch      mod_http2-2.0.42-1.amzn2023.0.1.x86_64
mod_lua-2.4.68-1.amzn2023.0.1.x86_64

Complete!
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-11-238 ~]$
```

i-03f191509e3878db6 (ELB-Server-1)
PublicIPs: 98.80.171.75 PrivateIPs: 172.31.11.238

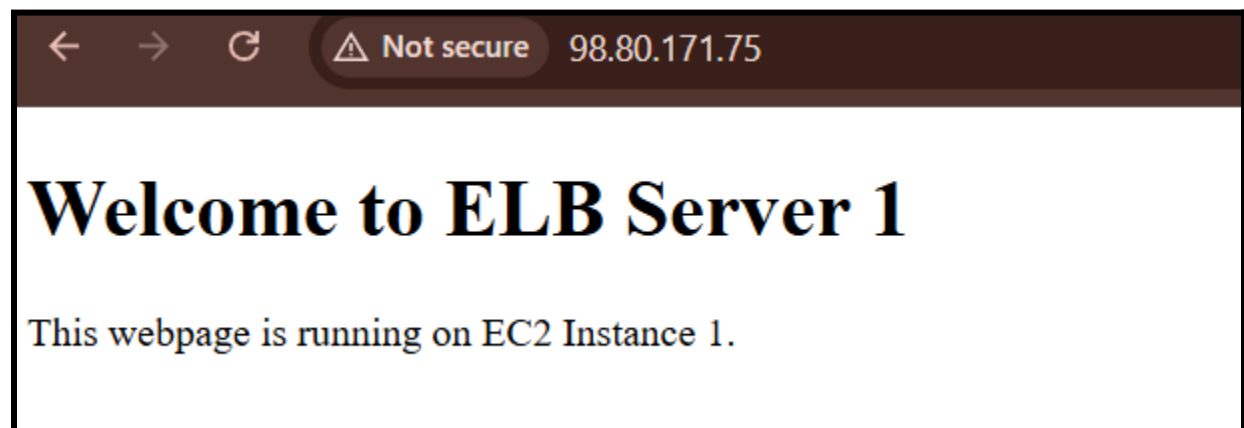
Created Server 1 webpage

Ran: `sudo nano /var/www/html/index.html`

Changed index.html to:

```
<!DOCTYPE html>
<html>
<head>
  <title>ELB Server 1</title>
</head>
<body>
  <h1>Welcome to ELB Server 1</h1>
  <p>This webpage is running on EC2 Instance 1.</p>
</body>
</html>
```

After installing the Apache webserver and editing, used the EC2 instance Public IP to check the webserver.



Configure ELB-Server-2

EC2 → Instances

ELB-Server-2 → Connect → EC2 Instance Connect

On terminal ran:

```
sudo dnf update -y
sudo dnf install httpd -y
sudo systemctl enable --now httpd
```

```

#_
~\#### Amazon Linux 2023
~~\#####\
~~\####|
~~\#/ https://aws.amazon.com/linux/amazon-linux-2023
~~V~' '->
~~~
~~.-.
~/m/'
[ec2-user@ip-172-31-10-12 ~]$ sudo dnf update -y
sudo dnf install httpd -y
sudo systemctl enable --now httpd

```

```

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64      apr-util-1.6.5-1.amzn2023.0.1.x86_64      apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64
apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64  generic-logos-httpd-18.0-12.amzn2023.0.3.noarch  httpd-2.4.68-1.amzn2023.0.1.x86_64
httpd-core-2.4.68-1.amzn2023.0.1.x86_64      httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch  httpd-tools-2.4.68-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64        mailcap-2.1.49-3.amzn2023.0.3.noarch        mod_http2-2.0.42-1.amzn2023.0.1.x86_64
mod_lua-2.4.68-1.amzn2023.0.1.x86_64

Complete!
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-10-12 ~]$

i-0bc9c5e2d385531cb (ELB-Server-2)
PublicIPs: 44.201.55.21  PrivateIPs: 172.31.10.12

```

Created Server 2 webpage

```
Ran: sudo nano /var/www/html/index.html
```

Changed index.html to:

```
<!DOCTYPE html>
<html>
<head>
  <title>ELB Server 1</title>
</head>
<body>
  <h1>Welcome to ELB Server 1</h1>
```

```
<p>This webpage is running on EC2 Instance 1.</p>
</body>
</html>
```

After installing the Apache webserver and editing, used the EC2 instance Public IP to check the webserver.



Configure ELB-Server-3

EC2 → Instances

ELB-Server-3 → Connect → EC2 Instance Connect

On terminal ran:

```
sudo dnf update -y
sudo dnf install httpd -y
sudo systemctl enable --now httpd
```



```

Installed:
apr-1.7.5-1.amzn2023.0.1.x86_64      apr-util-1.6.5-1.amzn2023.0.1.x86_64      apr-util-ldap-1.6.5-1.amzn2023.0.1.x86_64
apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64      generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      httpd-2.4.68-1.amzn2023.0.1.x86_64
httpd-core-2.4.68-1.amzn2023.0.1.x86_64      httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch      httpd-tools-2.4.68-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64      mailcap-2.1.49-3.amzn2023.0.3.noarch      mod_http2-2.0.42-1.amzn2023.0.1.x86_64
mod_lua-2.4.68-1.amzn2023.0.1.x86_64

Complete!
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service - /usr/lib/systemd/system/httpd.service.
ec2-user@ip-172-31-14-55 ~]$ sudo nano /var/www/html/index.html

```

Created Server 3 webpage

Ran: `sudo nano /var/www/html/index.html`

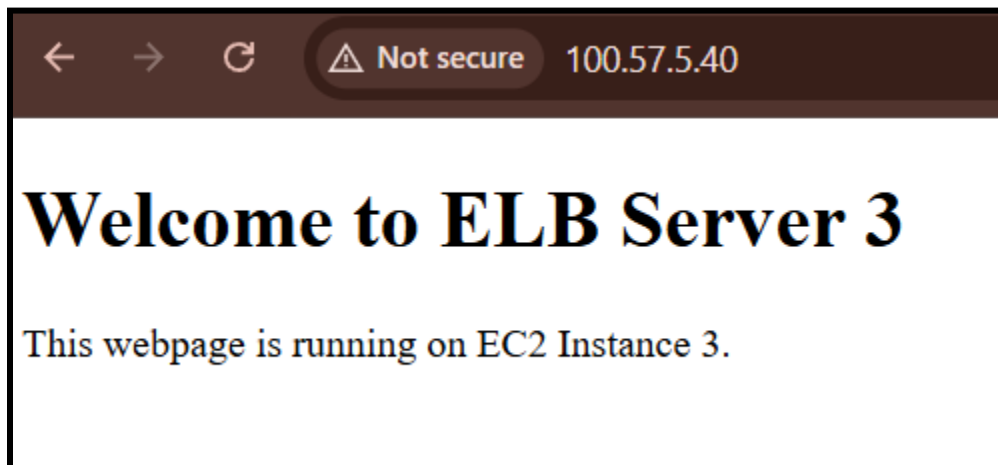
Changed index.html to:

```

<!DOCTYPE html>
<html>
<head>
  <title>ELB Server 1</title>
</head>
<body>
  <h1>Welcome to ELB Server 1</h1>
  <p>This webpage is running on EC2 Instance 1.</p>
</body>
</html>

```

After installing the Apache webserver and editing, used the EC2 instance Public IP to check the webserver.



3. To Create Classic Load Balancer

EC2 → Load Balancing → Load Balancers → Create Load Balancer

Selected Classic Load Balancer

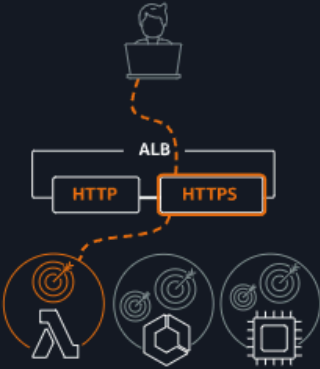
EC2 > Load balancers > Compare and select load balancer type

Compare and select load balancer type

A complete feature-by-feature comparison along with detailed highlights is also available. [Learn more](#)

Load balancer types

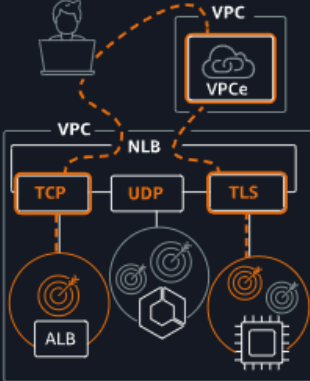
Application Load Balancer Info



Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

Create

Network Load Balancer Info



Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

Create

Gateway Load Balancer Info



Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

Create

► Classic Load Balancer - *previous generation*

Close

Configuration

Load Balancer name: XYZ-Classic-ELB

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.

XYZ-Classic-ELB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.

Since all three servers are in: us-east-1a, selected the VPC containing your three EC2 instances and the subnet for **us-east-1a**.

Network mapping

The load balancer routes traffic to targets in the selected subnets, and in accordance with your network settings.

VPC [Info](#)
Select the virtual private cloud (VPC) for your targets. Only VPCs with an internet gateway are available for selection. The selected VPC cannot be changed after the load balancer is created. When selecting a VPC for your load balancer, ensure each subnet has a CIDR block with at least a /27 bitmask and at least 8 free IP addresses.

[Learn more](#)

vpc-030682f2c586ad6b1 (default) [Refresh](#)

172.31.0.0/16

Availability Zones and subnets
Select at least one Availability Zone and one subnet for each zone. We recommend selecting at least two Availability Zones. The load balancer will route traffic only to targets that are not supported by the load balancer or the VPC are not available for selection.

☒ **us-east-1a (use1-az1)**

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-02791cbf0e8268fb9

172.31.0.0/20

Added and selected all three instances,

Add instances

Select EC2 instances to register to your load balancer. Requests will be routed to registered instances that meet the health check requirements. For maximum fault tolerance, we recommend maintaining approximately equivalent numbers of instances in each Availability Zone enabled for the load balancer. If demand on your instances changes, you can register or deregister instances without disrupting the flow of requests to your application. [Learn more](#)

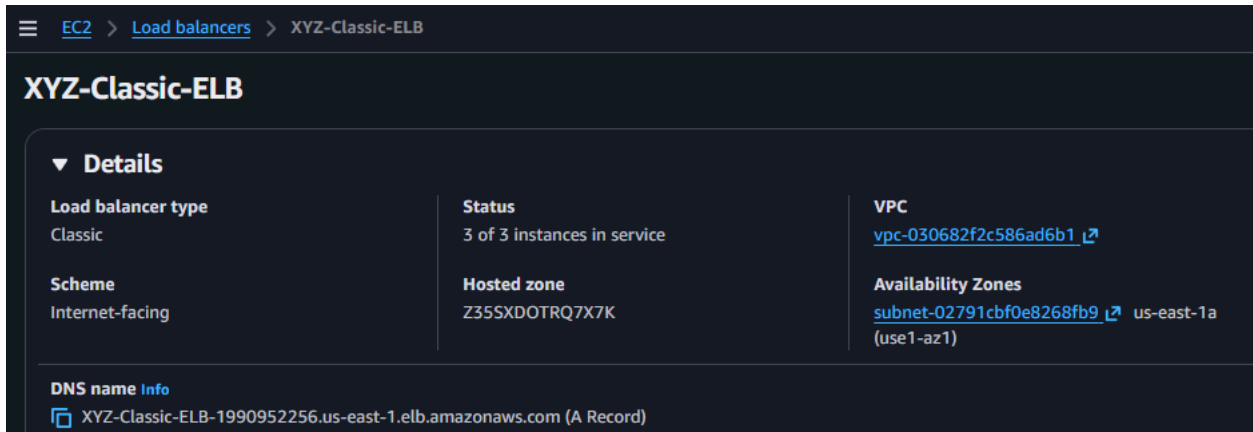
VPC
vpc-030682f2c586ad6b1

Available instances (3/3)

☒	Instance ID	Name	State	Security groups	Zone
☒	i-03f191509e3878db6	ELB-Server-1	Running	launch-wizard-36	us-east-1a
☒	i-0bc9c5e2d385531cb	ELB-Server-2	Running	launch-wizard-36	us-east-1a
☒	i-0420b5094ad8defef	ELB-Server-3	Running	launch-wizard-36	us-east-1a

[Cancel](#) [Confirm](#)

The Classic Load Balancer was created successfully.



The screenshot shows the AWS Management Console interface for a Classic Load Balancer. The breadcrumb navigation at the top indicates the path: EC2 > Load balancers > XYZ-Classic-ELB. The main heading is "XYZ-Classic-ELB". Below this, a "Details" section is expanded, showing a table of configuration details:

Load balancer type	Status	VPC
Classic	3 of 3 instances in service	vpc-030682f2c586ad6b1
Scheme	Hosted zone	Availability Zones
Internet-facing	Z35SXDOTRQ7X7K	subnet-02791cbf0e8268fb9 us-east-1a (use1-az1)

Below the table, the "DNS name" is listed as [XYZ-Classic-ELB-1990952256.us-east-1.elb.amazonaws.com](#) (A Record).

To Test the Classic Load Balancer:

Copied this DNS name from XYZ-Classic-ELB

<http://xyz-classic-elb-1990952256.us-east-1.elb.amazonaws.com/>

And pasted on browser and ran:



The screenshot shows a web browser window with the address bar displaying "xyz-classic-elb-1990952256.us-east-1.elb.amazonaws.com". The page content includes a large heading "Welcome to ELB Server 3" and a paragraph stating "This webpage is running on EC2 Instance 3."

The Classic Load Balancer is working successfully.

To Migrate to Application Load Balancer

1. Created the Target Group

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**

Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.

Suitable for: ALB NLB GWLB

☐ **IP addresses**

Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.

Suitable for: ALB NLB GWLB

☐ **Lambda function**

Supports load balancing to a single Lambda function. ALB required as traffic source.

Suitable for: ALB

☐ **Application Load Balancer**

Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.

Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.

XYZ-ALB-TG

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 10/32

Protocol

Protocol for communication between the load balancer and targets.

HTTP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ **IPv4**

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ **IPv6**

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-030682f2c586ad6b1
172.31.0.0/16

(default)



[Create VPC](#)

Selected all three EC2 as targets

Review targets

Targets (3) Remove all pending

Show only pending < 1 > ⚙️

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address
i-03f191509e3878db6	ELB-Server-1	80	Running	launch-wizard-36	us-east-1a	172.31.11.238
i-0bc9c5e2d385531cb	ELB-Server-2	80	Running	launch-wizard-36	us-east-1a	172.31.10.12
i-0420b5094ad8defef	ELB-Server-3	80	Running	launch-wizard-36	us-east-1a	172.31.14.55

Successfully created XYZ-ALB-TG

EC2 > Target groups > XYZ-ALB-TG

XYZ-ALB-TG

Actions

Details
[arn:aws:elasticloadbalancing:us-east-1:430776404757:targetgroup/XYZ-ALB-TG/ae044357b3ea649b](#)

Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1	VPC vpc-030682f2c586ad6b1
IP address type IPv4	Load balancer None associated		

3
Total targets

0
Healthy

0
Unhealthy

3
Unused

0
Initial

0
Draining

Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets Monitoring Health checks Attributes Tags

Registered targets (3) Info Anomaly mitigation: Not applicable Deregister Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

< 1 > ⚙️

☐	Instance ID	Name	Port	Zone	Health status	Health status details	Admini...	Overri...	La
☐	i-0bc9c5e2d385531cb	ELB-Server-2	80	us-east-1a (us...	Unused	Target group is not co...	-	-	Se
☐	i-03f191509e3878db6	ELB-Server-1	80	us-east-1a (us...	Unused	Target group is not co...	-	-	Se
☐	i-0420b5094ad8defef	ELB-Server-3	80	us-east-1a (us...	Unused	Target group is not co...	-	-	Se

2. Created Application Load Balancer

EC2 → Load Balancers → Create Load Balancer

Selected: Application Load Balancer. Clicked Create.

Configurations:

- Load Balancer Name: **XYZ-Application-ELB**
- Load Balancer Type: Application Load Balancer
- VPC: **vpc-030682f2c586ad6b1**
- Availability Zones/Subnets: Selected subnets in:
 - **us-east-1a**
 - **us-east-1b**
- Listener Protocol: HTTP, Listener Port: 80
- Forward traffic to **target group XYZ-ALB-TG**.

The Application Load Balancer successfully created.

The screenshot displays the AWS Management Console for the 'XYZ-Application-ELB'. At the top, a green banner confirms the successful creation of the load balancer. Below this, a blue banner introduces a new Capacity Utilization metric. The main section, titled 'XYZ-Application-ELB', contains a 'Details' tab with the following information:

Property	Value
Load balancer type	Application
Scheme	Internet-facing
Status	Provisioning
Hosted zone	Z35SXDOTRQ7X7K
VPC	vpc-030682f2c586ad6b1
Availability Zones	subnet-093a023aef10b9e26 (us-east-1b), subnet-02791cbf0e8268fb9 (us-east-1a)
Load balancer IP address type	IPv4
Date created	September 27, 2026, 12:26 (UTC+05:30)
Load balancer ARN	arn:aws:elasticloadbalancing:us-east-1:430776404757:loadbalancer/app/XYZ-Application-ELB/41a2a7f920f7cfc8
DNS name	XYZ-Application-ELB-321911158.us-east-1.elb.amazonaws.com (A Record)

Below the details, a navigation bar includes tabs for 'Listeners and rules', 'Network mapping', 'Resource map', 'Security', 'Monitoring', 'Integrations', 'Attributes', 'Capacity', and 'Tags'. The 'Listeners and rules' tab is active, showing a table with one listener:

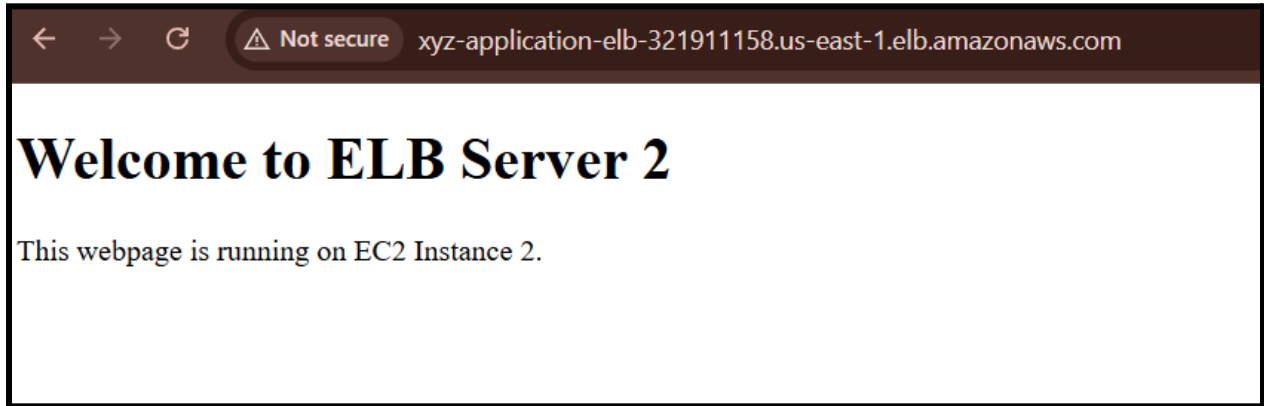
Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate
HTTP:80	Forward to target group XYZ-ALB-TG (100%) Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicable

3. Tested Application Load Balancer:

Copied the Application Load Balancer DNS name, pasted in the browser.

<http://xyz-application-elb-321911158.us-east-1.elb.amazonaws.com/>

The request was successfully forwarded to an EC2 instance, displaying:



Result:

Application Load Balancer was successfully created and configured to distribute HTTP traffic to the three registered EC2 web servers.